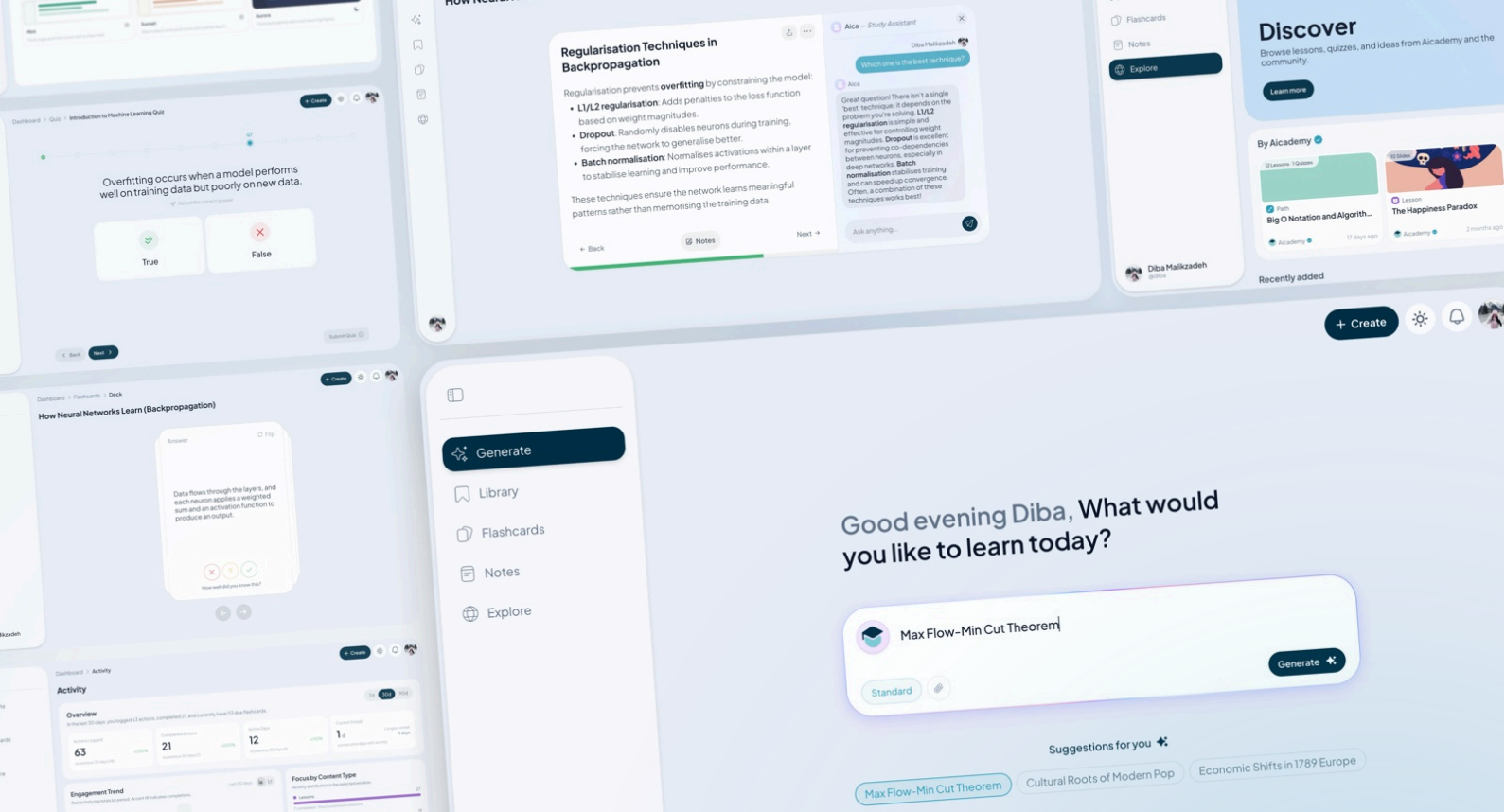




aicademy

PRODUCT SHOWCASE

# Aicademy

Everyone has the same AI. Not everyone gets the same out of it.

An AI-powered revision platform that turns any topic or uploaded documents into lessons, quizzes, flashcards and notes, with a tutor that already knows what the student is working on. Designed, built and shipped by one person.

LIVE AT

[useaicademy.com](https://useaicademy.com)

BUILT BY

Diba Malikzadeh

CONTACT

[contact@bydiba.dev](mailto:contact@bydiba.dev)

UPDATED

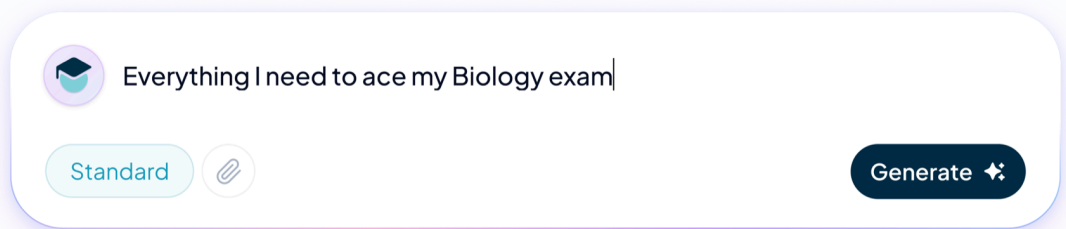
September 2026

# Why it exists

## THE GAP

### Getting something useful out of a chatbot is a skill, and it is not the one being examined.

The difference is not access. It is the prompting and the tool-switching it takes to turn an answer into something worth revising from, and nothing that comes back is saved anyway. **Aicademy exists so that none of that is the student's job.**



The screenshot shows a chat input box with a purple header bar. Inside the box, there is a purple circular icon with a white graduation cap. To the right of the icon, the text "Everything I need to ace my Biology exam" is entered. Below the text, there are two buttons: a light blue button labeled "Standard" and a light blue button with a paperclip icon. To the right of these buttons is a dark blue button labeled "Generate" with a white star icon.

## THE STORY

Lecture slides never worked for me, so I used AI to turn whatever I was studying into something I could actually revise from. People started asking for what I had made, even though they could have asked for it themselves. The gap was knowing how to prompt. Noticing that is what gave me the idea for Aicademy.

When my final year started I asked to carry on with it as my project. It was graded a First in March, and **about three quarters of the work has happened since**. The people testing it kept asking whether the site would stay up, so it did.

I used AI the way I would use any other tool: to explore options, to move faster through repetitive changes, and to argue with. Every decision in here is one I made and can defend.

## CURRENTLY

694

Lessons written to an exam specification

39

Diagram types, each with its own custom renderer

12

Subjects across AQA, Edexcel and OCR

128

API routes in one TypeScript codebase

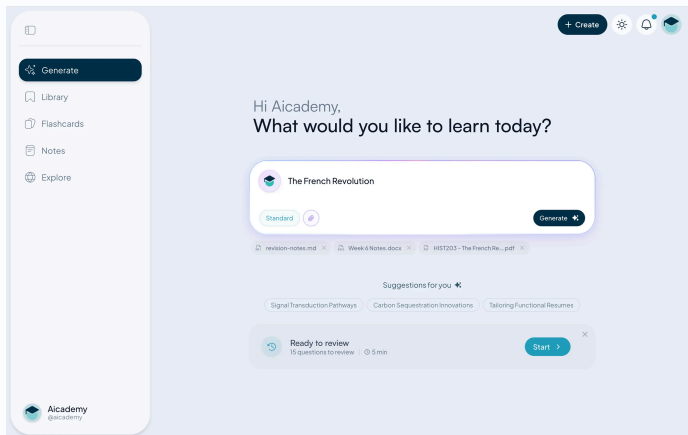
6

Settings that change what the AI writes

Live

In production since August 2025

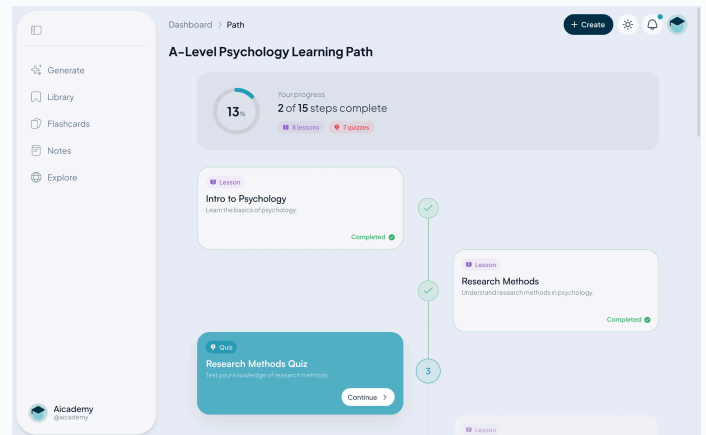
# What's inside



01

## Start from anything

Enter a topic or upload material, and optionally choose a difficulty level. Aicademy reads the request and decides what fits best: a single lesson, a learning path, a quiz or a flashcard deck.



02

## Learning paths

A whole subject broken into lessons and quizzes, planned in one pass, so every lesson builds on the last. Each step unlocks the next — there is always a clear next move instead of a syllabus to stare at.

## Meet the Halogens: Group 7 Overview

Group 7 elements — fluorine, chlorine, bromine, iodine, astatine, tennessine — are called the **halogens**. They sit one column from the right of the periodic table and share key behaviours because they have seven electrons in their outer shell.

A halogen atom tends to gain one electron to form a stable octet.

- Common oxidation states: **-1** in halide ions (e.g.,  $\text{Cl}^-$ ), sometimes +1, +3, +5, +7 in compounds.
- Diatomic molecules at standard conditions: **F<sub>2</sub>, Cl<sub>2</sub>, Br<sub>2</sub>, I<sub>2</sub>**.
- Typical colours and states: **F<sub>2</sub>** pale yellow gas, **Cl<sub>2</sub>** green gas, **Br<sub>2</sub>** red-brown liquid, **I<sub>2</sub>** grey-black solid.

Why this matters: the seven valence electrons drive their **reactivity**, especially in redox reactions and salt formation (e.g.,  $2\text{NaCl}$ ).

Group 7 — the halogens

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|----|
| 1  | H  | 2  | He |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |
| 3  | Li | 4  | Be | 5  | B  | 6  | C  | 7  | N  | 8  | O  | 9  | F  | 10 | Ne |    |    |    |    |    |    |    |    |    |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |
| 11 | Na | 12 | Mg | 13 | Al | 14 | Si | 15 | P  | 16 | S  | 17 | Cl | 18 | Ar |    |    |    |    |    |    |    |    |    |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |
| 19 | K  | 20 | Ca | 21 | Sc | 22 | Ti | 23 | V  | 24 | Cr | 25 | Mn | 26 | Fe | 27 | Co | 28 | Ni | 29 | Cu | 30 | Zn | 31 | Ga | 32  | Ge | 33  | As | 34  | Se | 35  | Br | 36  | Kr |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |
| 37 | Rb | 38 | Sr | 39 | Y  | 40 | Zr | 41 | Nb | 42 | Mo | 43 | Tc | 44 | Ru | 45 | Rh | 46 | Pd | 47 | Ag | 48 | Cd | 49 | In | 50  | Sn | 51  | Sb | 52  | Te | 53  | I  | 54  | Xe |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |     |    |
| 55 | Cs | 56 | Ba | 57 | La | 58 | Ce | 59 | Pr | 60 | Nd | 61 | Pm | 62 | Sm | 63 | Eu | 64 | Gd | 65 | Tb | 66 | Dy | 67 | Ho | 68  | Er | 69  | Tm | 70  | Yb | 71  | Lu | 72  | Hf | 73  | Ta | 74  | W  | 75  | Re | 76  | Os | 77  | Ir | 78  | Pt | 79  | Au | 80  | Hg | 81  | Tl | 82  | Pb | 83  | Bi | 84  | Po | 85  | At | 86  | Rn |
| 87 | Fr | 88 | Ra | 89 | Ac | 90 | Th | 91 | Pa | 92 | U  | 93 | Np | 94 | Pu | 95 | Am | 96 | Cm | 97 | Bk | 98 | Cf | 99 | Es | 100 | Fm | 101 | Md | 102 | No | 103 | Lr | 104 | Rf | 105 | Db | 106 | Sg | 107 | Bh | 108 | Hs | 109 | Mt | 110 | Ds | 111 | Rg | 112 | Cn | 113 | Nh | 114 | Fl | 115 | Mc | 116 | Lv | 117 | Ts | 118 | Og |

### Notes

Group 7 (Halogens): properties & r...

configuration  $ns^2 np^5$ ; exist as **diatomic** molecules **F<sub>2</sub>, Cl<sub>2</sub>, Br<sub>2</sub>, I<sub>2</sub>**

- Common oxidation states** -1 in halides (e.g.,  $\text{Cl}^-$ ); also +1, +3, +5, +7 in oxohalogen compounds
- Key physicals** at room conditions: F<sub>2</sub> pale yellow gas, Cl<sub>2</sub> green gas, Br<sub>2</sub> red-brown liquid, I<sub>2</sub> grey-black solid (purple vapour)

### Ask Aica

Aicademy

In the second bullet, why do the colours get darker as you go down the group?

Aica

Going down the group the molecules get bigger and have more electrons, so they absorb more visible light. The more they absorb, the deeper the colour: pale yellow F<sub>2</sub>, green Cl<sub>2</sub>, red-brown Br<sub>2</sub>, then almost black I<sub>2</sub>.

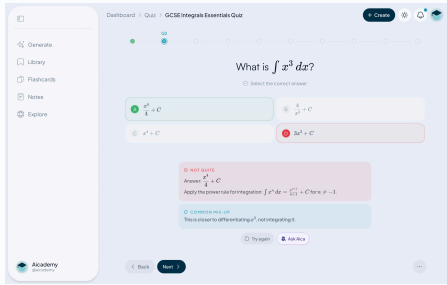
Ask anything...

03

## Lessons

Each lesson is a set of slides rather than an essay, one idea at a time, with tables, code and diagrams built in. The tutor and notes sit on the same page, so nothing pulls the student into another tab.

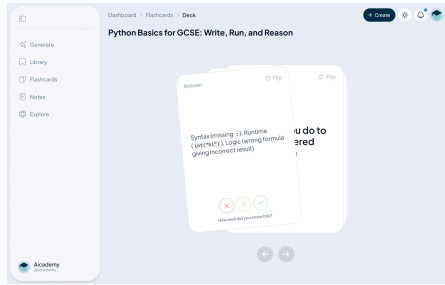
# What's inside, continued



04

## Quizzes

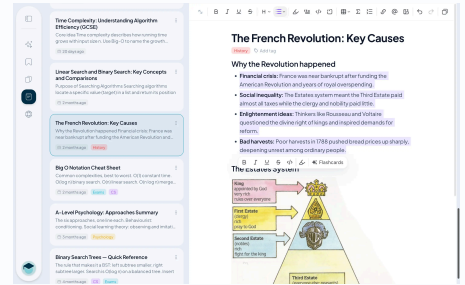
Five question types, two modes and a tutor built in. Every answer is explained, with the common misconceptions flagged.



05

## Flashcards

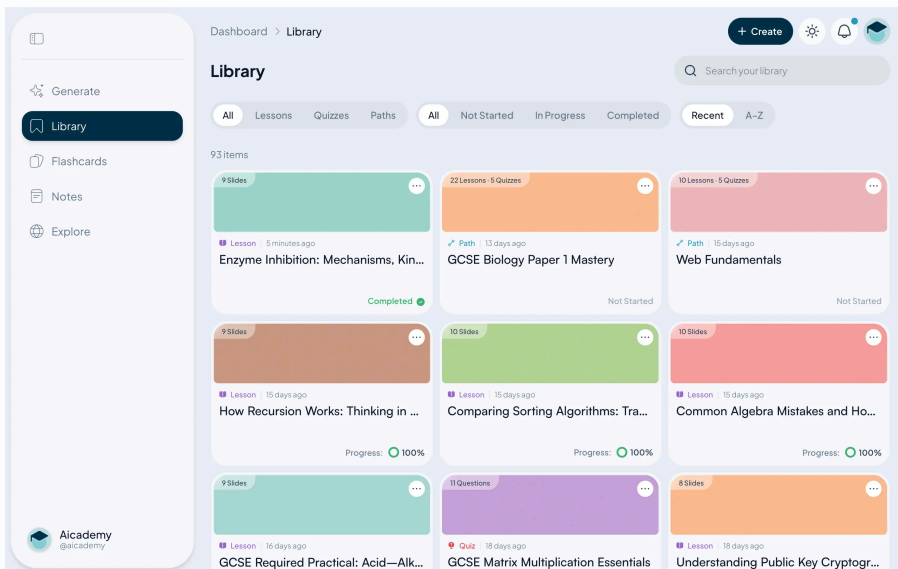
Scheduling follows SM-2 — a card recalled easily drops out of rotation, and one missed keeps coming back until it sticks.



06

## Notes

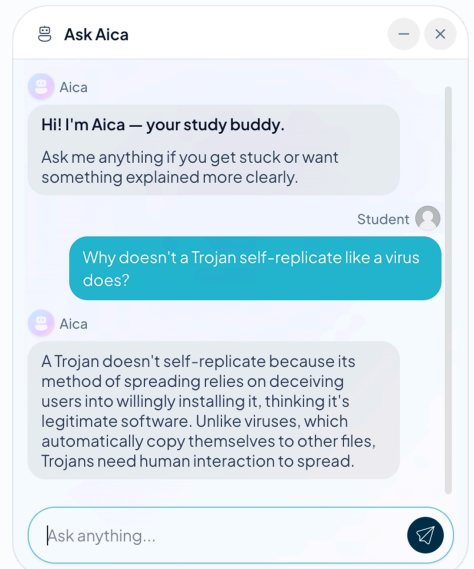
A Tiptap editor with headings, tables, code and LaTeX. Highlight a passage and turn it into flashcards. Mention anything using @.



07

## The library

Everything a student makes ends up here, with its progress attached and searchable: lessons, quizzes and paths. This is the part a chat window cannot do, and the reason the whole thing is relational rather than a transcript.



08

## Aica, the tutor

Sits inside lessons and quizzes, knows the slide the student is on and how they learn, and can suggest edits to it.

### THE COMPARISON

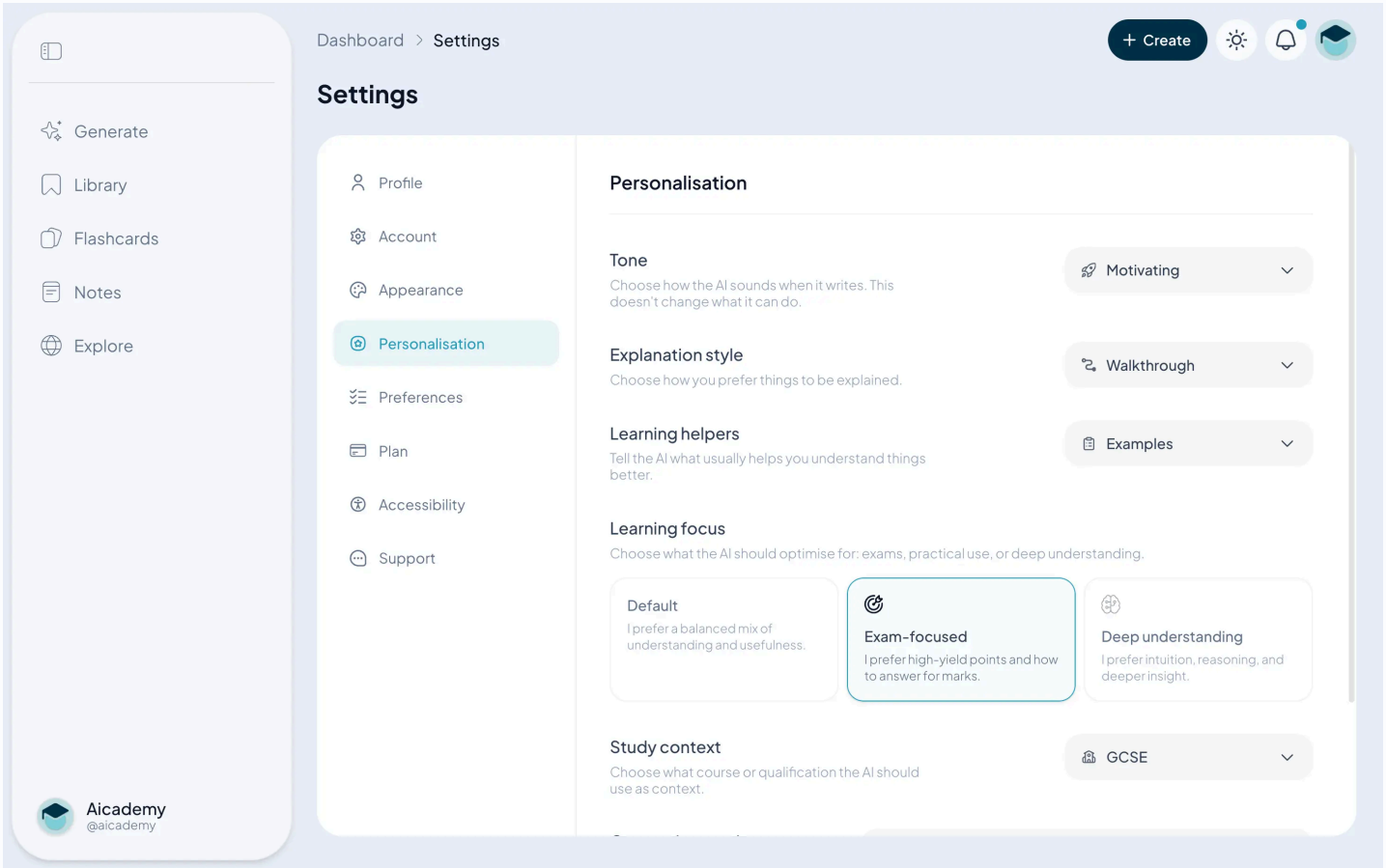
ChatGPT explains things well, and that is the problem. A clear explanation feels like learning: asked to predict what they will remember, students turn out to be poor judges of what they can actually recall later. So nothing here stops at the explanation.

The full comparison: [useaicademy.com/blog/aicademy-vs-chatgpt](https://useaicademy.com/blog/aicademy-vs-chatgpt)



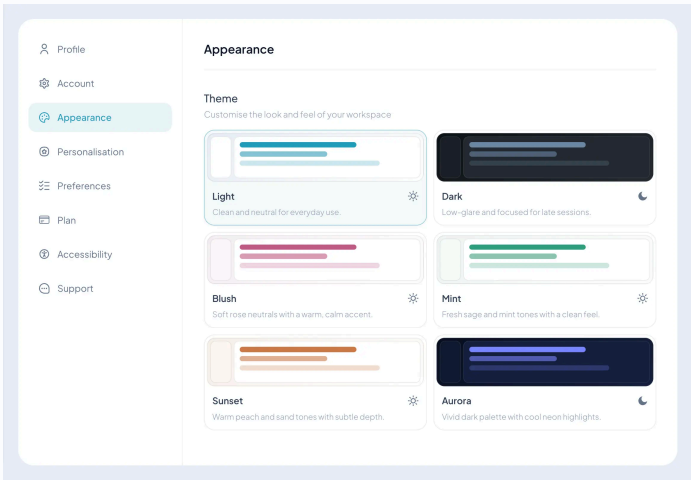
# Built around the student

A good teacher does not explain the same way to everyone. "Built around you" began as that idea, and it did not stay in the writing.



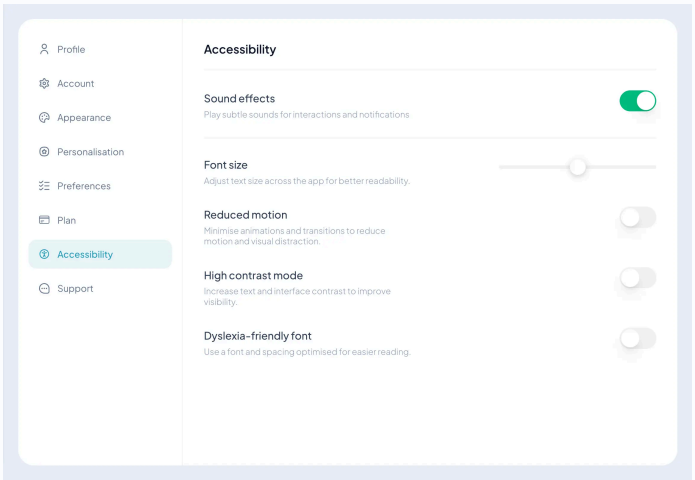
## Personalisation

Six settings shape how everything is written: tone, explanation style, learning helpers, learning focus, study context and free-text instructions. A few questions at the start set them up, so the first lesson already fits.



## Appearance

Six themes to choose from, so the workspace suits whoever is using it: Light, Dark, Blush, Mint, Sunset and Aurora.



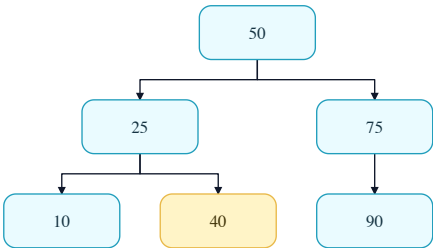
## Accessibility

Reduced motion, high contrast, dyslexia-friendly type and font scaling, and everything underneath that has no switch to show.

# The model is never allowed to draw

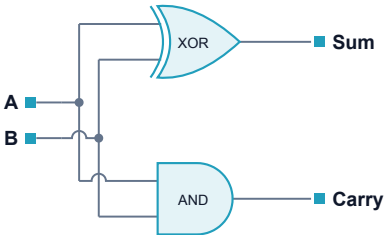
A lesson needs diagrams, and there was no way to get them: generated images were unusable and no library covered the syllabus. So I took the idea behind Mermaid, where a model writes a fixed notation and something else draws it, and built it for school subjects: **39 schemas so far**, each with its own renderer. The model supplies values, never markup, and a block that fails validation is dropped rather than shown.

SOME EXAMPLES



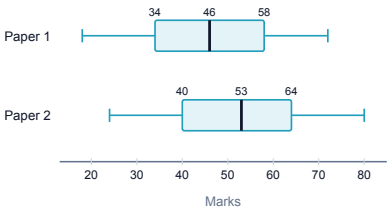
Binary search tree

DIAGRAM · TREE



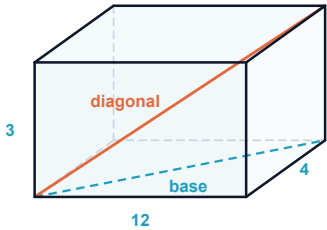
Half adder

CIRCUIT



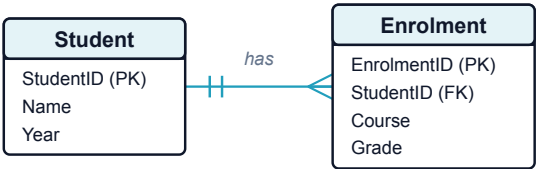
Marks by paper

STAT · BOXPLOT



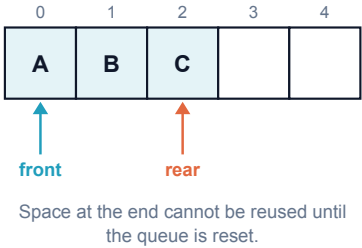
Space diagonal

SOLID3D



One to many

ER



Linear queue

QUEUE

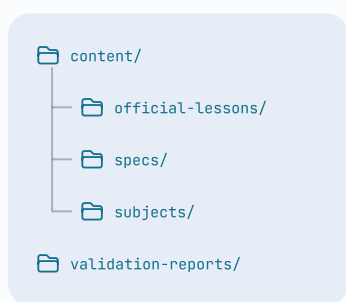
# How the free lessons are made

## MODEL OUTPUT

The first draft is written by AI and I say so on the site, because pretending otherwise would serve nobody. The rest is a separation of duties: the skill that writes a lesson is never the skill that approves it, because a model asked to judge its own work will favour it. Four Claude Code skills, each running with its own scoped tool permissions.

The published version: [useaacademy.com/about/our-content](https://useaacademy.com/about/our-content)

## THE REPO



## Claude Code



ONE SKILL LOADED AT A TIME

## .CLAUDE/SKILLS

/aicademy-lesson-author

/aicademy-lesson-validator

/aicademy-curriculum-auditor

/aicademy-asset-validator

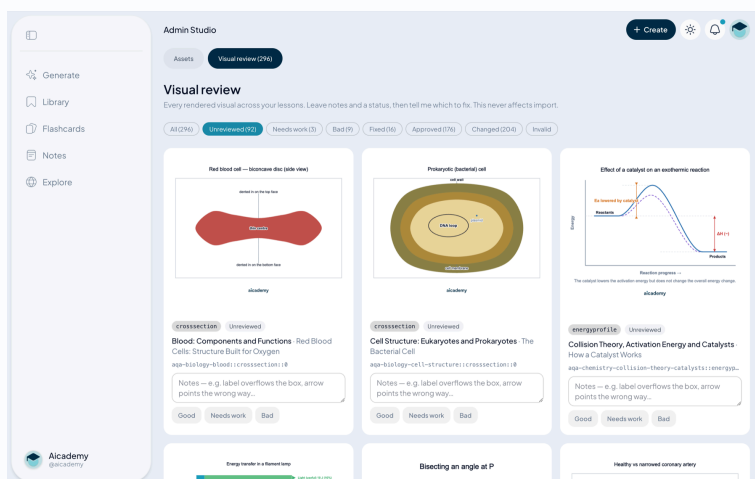
## SPEC - LOCKED

Every lesson is locked to one exam board and checked against that board's own specification document, not written from general knowledge and hoped to line up. Anything falling just outside the spec is labelled as extra context. All of that review sits behind **694 lessons that need no account to read**.

These lessons are published at [useaacademy.com/explore](https://useaacademy.com/explore)

## HUMAN REVIEWED

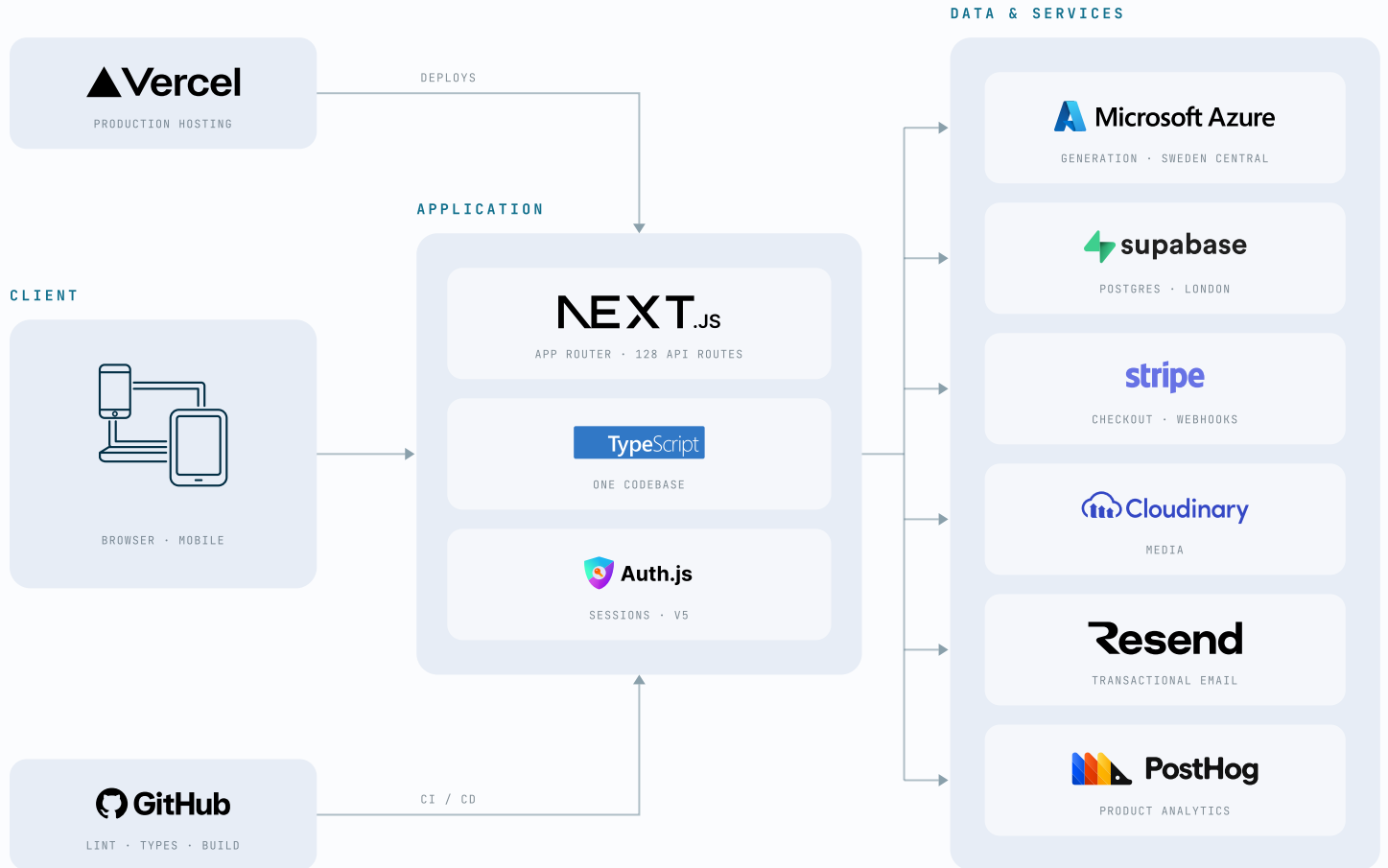
A schema proves a block is **well-formed, not well-drawn**. So I built a studio where every diagram renders on one page, and I mark each with a note on how it is drawn. Individually those notes are complaints; together they are a list of failure modes, and one fix lands across every diagram that shares one.



# Under the hood

## THE SHAPE

Aicademy is built on Next.js and runs on Vercel, with a managed service behind everything else. Every box below is called in production today.



## WHERE IT RUNS

Generation goes through Azure OpenAI, Microsoft's enterprise service rather than the consumer product. The resource sits in **Sweden Central**, prompts stay inside the EU data zone, and accounts, lessons, progress and tutor history live in a single Supabase Postgres database in London. Every sub-processor is documented with what it sees and how long it keeps it, including the Microsoft abuse-monitoring exemption I have not applied for.

Advice from a school sent me to check what the enterprise service guarantees, and to build the parts it does not. Prompts are not used to train a model; the upload screen asks for study material and nothing else, and the tutor will not repeat personal details about other people back to a student. Accounts require confirming the account holder is 13 or over, and because the material is GCSE and A-level, **most users are between 13 and 17**.

Documented in full at [useaicademy.com/ai-and-data](https://useaicademy.com/ai-and-data)

# What it cost

## DOCUMENT UPLOAD

An upload is read once and rewritten into numbered sections that can be taught from on their own, and each is then assigned to the lessons and quizzes that need it. That is retrieval-augmented generation with **the retrieval done by reading rather than by similarity search**. It works because the path is planned before anything is written; adaptive paths are the change that would make a vector store worth reconsidering.

## QUIZ MARKING

Three of the five question types have unambiguous answers. Fill in the blank is matched against a list of accepted forms generated with the question, covering spellings, synonyms and the usual misspellings. Short answers run through the same marker and are the weak point: **one typed line has too many right shapes for a fixed list to hold**. That is where an AI marker starts to make sense, especially once quizzes become practice papers.

## USER TESTING

I ran two anonymised rounds of user testing for the final project: an initial one, and another after I had fixed the problems it turned up.

| MEASURE                         | ROUND 1 · N=6 | ROUND 2 · N=12 |
|---------------------------------|---------------|----------------|
| Ease of use                     | 4.8           | 5.0            |
| Usefulness                      | 4.8           | 4.9            |
| Trust in AI-generated content   | 4.7           | 4.6            |
| Likelihood of reuse             | 4.3           | 4.8            |
| Confidence in the topic, before | 1.5           | 2.8            |
| Confidence in the topic, after  | 4.5           | 4.5            |

Twelve people self-reporting is not evidence that the product is good. What it shows is that the changes landed: the seven who had seen the earlier version rated it improved, and the written answers kept saying the same thing, that it worked as **one tool rather than several**.

## ORGANIC SEARCH

Every lesson is indexed and impressions have grown from nothing in three months. The page that ranks first gets **thousands of impressions and no clicks**, because Google's AI overview answers the search in the results. What arrives is mostly British, and a quarter of the people who click anything make an account. Bing runs at five times Google's click rate on a fraction of the impressions, which fits school machines where Edge is the default.

## WHAT I'D FIX

Mobile converts at about a third of the desktop rate, and that is the first thing I would fix. The rest is **what one person can defer and a team cannot**: schema changes are made in Supabase rather than versioned with the code, there is no code review beyond my own, and nothing alerts me when something breaks in production.



# Close

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Quizzes and flashcards aren't decoration. Testing yourself improves retention more than re-reading does [1], spacing that testing out works better than doing it in one go [2], and students do better when they can see and steer their own plan [3]. Those three findings are why every lesson can become a quiz, why spaced repetition is built in, and why the product has learning paths.

- [1] Roediger, H. L. & Karpicke, J. D. (2006). Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention. *Psychological Science*, 17(3), 249–255.
- [2] Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T. & Rohrer, D. (2006). Distributed Practice in Verbal Recall Tasks: A Review and Quantitative Synthesis. *Psychological Bulletin*, 132(3), 354–380.
- [3] Zimmerman, B. J. (2002). Becoming a Self-Regulated Learner: An Overview. *Theory Into Practice*, 41(2), 64–70.

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## FIND IT

### PRODUCT

[useaacademy.com](https://useaacademy.com)

### LINKEDIN

[linkedin.com/company/useaacademy](https://linkedin.com/company/useaacademy)

### REPOSITORY

[github.com/dmalikzadeh/aicademy](https://github.com/dmalikzadeh/aicademy)

### PORTFOLIO

[bydiba.dev](https://bydiba.dev)

## WHO MADE IT

Diba Malikzadeh. BSc Computer Science, first class, Birmingham 2026. Looking for graduate software engineering work in the UK.

[contact@bydiba.dev](mailto:contact@bydiba.dev)



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